

Mathematics - 7

Differential Equation

* e.g. $\sin x$, $\log \frac{dy}{dx}$, $\sin \frac{dy}{dx}$ - degree is not defined.

formation of diff. equation

First arbitrary const. 3rd order - 3 diff. eq. are formed by differentiation w.r.t. x

Q. $y = Ax + A^2 \rightarrow (1)$
 - Sol. differentiate eqn w.r.t. x
 $\frac{dy}{dx} = A$

now form equation 1

$y = \frac{dy}{dx}x + \left(\frac{dy}{dx}\right)^2$
 homogeneous

$\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx}x - y = 0$

Order = 1
 degree = 2

Q. $y^2 = Ax^2 + Bx + C$

Sol. diff. eqn w.r.t. x

$2y \frac{dy}{dy} = 2x \cdot A + B + 0$

Again differentiate

$2y \frac{dy}{dx} + 2 \frac{dy}{dx}x \frac{dy}{dx} = 2A$

$2y \frac{dy}{dx} + 2x \frac{dy^2}{dx} = 2A$

$y \frac{d^3y}{dx^3} + 3 \frac{dy}{dx} \times \frac{d^2y}{dx^2} = 0$

Differential equation in variable separation form:

If diff. eqn can be put in the form $\frac{dy}{dx} = f(x)g(y)$ separate it

Sol. $\frac{dy}{dx} = f(x)g(y) = \int g(y)dy + C$

Q. $\frac{dy}{dx} = (4x + y + 1)^2$

put $4x + y + 1 = v$

differentiate by $\frac{dy}{dx}$

$\frac{dy}{dx} = \frac{dv}{dx} + 4$

$\frac{dy}{dx} = \frac{dv}{dx} - 4$

putting value of $\frac{dy}{dx}$ in eq

$\frac{dv}{dx} - 4 = v^2 \Rightarrow \frac{dv}{dx} = v^2 + 4$

$\int \frac{dv}{(v^2+4)} = \int dx$

$\frac{1}{2} \tan^{-1} \frac{v}{2} = x + C$

$\left[\frac{1}{2} \tan^{-1} \left(\frac{4x+y+1}{2} \right) = x + C \right]$

Homogeneous diff. eqn

$\frac{dy}{dx} = \frac{3xy + y^2}{3x^2 + xy}$ put $y = vx$

$\frac{dy}{dx} = v + x \frac{dv}{dx}$

x & y have same degree

Equation Reducible to Homogeneous

Given / non-homogeneous dependent equation
 $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$

Case 1

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

then substitute
 $x = X+h$
 $y = Y+k$

Solve
 $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3} \rightarrow (1)$

where $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, put $x = X+h$

from eqn (1)
 $dy = dx$
 $\frac{dy}{dx} = 1$
 $y = X+h$

put $Y = Y-h$
 $\frac{dY}{dX} = \frac{X+h+2(Y-h)-3}{2(X+h)+(Y-h)-3}$

from eq we get
 $\frac{dY}{dX} = \frac{X+2Y-3}{2X+Y-3}$

we choose h & k such that
 $h+2k-3=0$
 $2h+k-3=0$

$h=1$ & $k=1$

$\therefore \frac{dY}{dX} = \frac{X+2Y-3}{2X+Y-3}$

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So

$$\frac{(x-1+y-1)}{(x+y-2)} = \frac{(x-1-y+1)^2}{(x-y)^2 c^2}$$

Case II

$$\left[\frac{dy}{dx} = \frac{a_1 y + b_1}{a_2 y + b_2} \right]$$

e.g. $\frac{dy}{dx} = \frac{x+2y-1}{x+2y+1}$

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Then we substitute this eqⁿ (1) that's how we can have to solve

Let, $x+2y=2$

$1+2 \frac{dy}{dx} = \frac{dz}{dx}$

Differentiation formula :-

- 1) $\frac{d}{dx} x^n = nx^{n-1}$
- 2) $\frac{d}{dx} x = 1$
- 3) $\frac{d}{dx} c = 0$
- 4) $\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$
- 5) $\frac{d}{dx} \left(\frac{1}{x}\right) = -\frac{1}{x^2}$
- 6) $\frac{d}{dx} e^x = e^x$
- 7) $\frac{d}{dx} \log x = \frac{1}{x}$
- 8) $\frac{d}{dx} a^x = a^x \log a$
- 9) $\frac{d}{dx} \sin x = \cos x$
- 10) $\frac{d}{dx} \cos x = -\sin x$
- 11) $\frac{d}{dx} \tan x = \sec^2 x$
- 12) $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$
- 13) $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$
- 14) $\frac{d}{dx} \sec x = \sec x \tan x$
- 15) $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$
- 16) $\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$
- 17) $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$
- 18) $\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$
- 19) $\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$
- 20) $\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$
- 21) $\frac{d}{dx} (u.v) = u \frac{dv}{dx} + v \frac{du}{dx}$
- 22) $\frac{d}{dx} \left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Integration formula

- 1) $\int x^n dx = \frac{x^{n+1}}{n+1} + c$
- 2) $\int x dx = \frac{x^2}{2} + c$
- 3) $\int \frac{1}{x} dx = \log x + c$
- 4) $\int \frac{1}{x^2} dx = -\frac{1}{x} + c$
- 5) $\int e^x dx = e^x + c$
- 6) $\int \sin x dx = -\cos x + c$
- 7) $\int \cos x dx = \sin x + c$
- 8) $\int \tan x dx = \log |\sec x| + c$
- 9) $\int \cot x dx = \log |\sin x| + c$
- 10) $\int \sec x dx = \log |\sec x + \tan x| + c$
- 11) $\int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + c$
- 12) $\int \sec^2 x dx = \tan x + c$
- 13) $\int \operatorname{cosec}^2 x dx = -\cot x + c$
- 14) $\int \sec x \tan x dx = \sec x + c$
- 15) $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$
- 16) $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$
- 17) $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$
- 18) $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left| \frac{x}{a} \right| + c$
- 19) $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + c$
- 20) $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log |x + \sqrt{x^2 + a^2}| + c$

$$21) \int \frac{1}{\sqrt{x^2 - a^2}} dx = \log|x + \sqrt{x^2 - a^2}| + C$$

$$22) \int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1}(u) + C = -\cos^{-1}(u)$$

$$23) \int \frac{1}{1+u^2} du = \tan^{-1}(u) + C = -\cot^{-1}(u)$$

$$24) \int \frac{1}{u\sqrt{u^2-1}} du = \sec^{-1}(u) + C = -\operatorname{cosec}^{-1}(u)$$

$$25) \int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{u}{a}\right)$$

$$26) \int \sqrt{x^2 - a^2} = \frac{u}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2 - a^2}|$$

$$27) \int \sqrt{u^2 + a^2} du = \frac{u}{2} \sqrt{u^2 + a^2} + \frac{a^2}{2} \log|u + \sqrt{u^2 + a^2}|$$

$$28) \int e^{au} \sin bu dx = \frac{e^{au}}{a^2 + b^2} (a \sin bu - b \cos bu) + C$$

$$29) \int e^u (f(u) + f'(u)) du = e^u f(u)$$

$$30) \int (u.v) du = u \int v dv - \int \frac{du}{dt} \cdot \int v dv$$

Linear diff. equations

$$\boxed{\frac{dy}{dx} + Py = Q}$$

where P & Q are function of x or constant.

Integrating factor $\Rightarrow e^{\int P dx}$

Solution written $\Rightarrow \boxed{y \cdot I.F. = \int Q \cdot I.F. dx + C}$

Q $(x+1) \frac{dy}{dx} - y = e^x (x+1)^2$

Sol $\frac{(x+1) dy}{(x+1) dx} - \frac{y}{(x+1)} = e^x \frac{(x+1)}{(x+1)}$

$$\frac{dy}{dx} - \frac{y}{(x+1)} = e^x (x+1)$$

this is a form of linear eqn.

$\therefore I.F. = e^{\int -\frac{1}{(x+1)} dx}$

$$\Rightarrow e^{-\log(x+1)}$$

$$\Rightarrow e^{\log(x+1)^{-1}}$$

$$\Rightarrow \boxed{\frac{1}{(x+1)} \Rightarrow I.F.}$$

Solution $y \left(\frac{1}{x+1} \right) = \int \frac{1}{x+1} e^x (x+1) dx$

Ans $\boxed{\frac{y}{(x+1)} = e^x + C}$

Equation reducible to linear form

$$\boxed{\frac{dy}{dx} + Py = Qy^n} \rightarrow \text{Euler}$$

$$\frac{1}{y^n} \frac{dy}{dx} + \frac{P}{y^{n-1}} = Q$$

let $y^{n-1} = Z \Rightarrow$

$$\Rightarrow \frac{1}{y^n} \frac{dy}{dx} + ZP = Q$$

$$\Rightarrow \frac{1}{y^{n-1}} = Z \text{ differentiate}$$

$$+ -(n-1)y^{-n} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow y^{-n} \frac{dy}{dx} = \frac{-1}{(n-1)} \frac{dz}{dx}$$

Q $x \frac{dy}{dx} + y \log y = x y e^x$

Sol $\frac{x dy}{x dx} + \frac{y \log y}{x} = \frac{x}{x} y e^x$

$$\frac{dy}{dx} + \frac{y \log y}{x} = y e^x$$

divided by y

$$\frac{1}{y} \frac{dy}{dx} + \frac{1}{x} \log y = e^x \quad \text{--- (i)}$$

let $\log y = z$

differentiate both side

$$\frac{1}{y} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dz}{dx} + \frac{z}{x} = e^x \quad \text{putting in (i)}$$

$I.F. \Rightarrow e^{\int \frac{1}{x} dx} \Rightarrow e^{\log x} \Rightarrow x$

Solution $\Rightarrow y z = \int e^x \cdot x$

$$zy = x e^x - e^x + C$$

$$x \log y = x e^x - e^x + C \quad \text{Ans}$$

$$\Rightarrow \frac{dy}{dx} - \frac{\tan y}{(1+x)} = (1+x) e^x \sec y$$

Sol: dividing by $\sec y$

$$+ \frac{1}{\sec y} \frac{dy}{dx} - \frac{\tan y}{\sec y (1+x)} = (1+x) e^x$$

$$+ \frac{1}{\sec y} \cos y \frac{dy}{dx} - \frac{\sin y}{(1+x)} = (1+x) e^x \rightarrow (i)$$

$$\rightarrow \text{Let } \sin y = z$$

$$\Rightarrow \sin y \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \cos y \frac{dy}{dx} = \frac{dz}{dx}$$

$\Rightarrow$ Substitute value in eq (i)

$$\Rightarrow \frac{dz}{dx} - \frac{z}{(1+x)} = (1+x) e^x$$

$$\Rightarrow \text{I.F} = e^{\int -\frac{1}{1+x} \frac{dy}{dx}}$$

$$= e^{\log(1+x)^{-1}}$$

$$\Rightarrow \text{I.F} = \frac{1}{(1+x)}$$

$$\Rightarrow \text{Solution } \text{I.F} \cdot \frac{dz}{dx} = \int 0 \cdot \text{I.F} + C$$

$$\Rightarrow \frac{z}{(1+x)} = \frac{(1+x) e^x}{(1+x)} + C$$

$$\text{Ans} \quad \frac{\sin y}{(1+x)} = e^x + C$$

H.W

$$\Rightarrow x^2 dy + y(x+y) dx = 0$$

$$\text{Ans} \Rightarrow \frac{1}{xy} = -\frac{1}{2x^2} + C$$

$$2) \frac{dy}{dx} + y \cot x = \cos x$$

$$\text{Ans} \Rightarrow y \sin x = \frac{1}{2} \frac{\cos 2x}{2} + C$$

$$3) (x^2+1) \frac{dy}{dx} + 2xy = x^2$$

$$\text{Ans} \Rightarrow y(x^2+1) = \frac{x^3}{3} + C$$

Exact diff. eqn

$$Mdx + Ndy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\text{Solution} \Rightarrow \int M dx + \int N dy$$

y constant (those terms not containing x)

$$= C$$

$$\Rightarrow (5x^4 + 3x^2y^2 - 2xy^3) dx + (2x^3y - 3x^2y^2 - 5y^4) dy = 0$$

Sol: this eqn is in the form

$$Mdx + Ndy = 0$$

where

$$M = (5x^4 + 3x^2y^2 - 2xy^3)$$

$$N = (2x^3y - 3x^2y^2 - 5y^4)$$

$\Rightarrow$

H.W Solution

$$1) x^2 dy + y(x+y) dx = 0$$

$$\text{sol } x^2 \frac{dy}{dx} + yx + y^2 = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} + \frac{y^2}{x^2} = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = -\frac{y^2}{x^2}$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} y = -\frac{1}{x^2}$$

$$\text{let } \frac{1}{y^2} = z \Rightarrow -\frac{1}{y^2} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow -\frac{dz}{dx} + \frac{z}{x} = -\frac{1}{x^2}$$

$$\Rightarrow \frac{dz}{dx} - \frac{z}{x} = \frac{1}{x^2}$$

$$I.F = \int e^{\int -\frac{1}{x} dx}$$

$$I.F = e^{\log x^{-1}} \Rightarrow \frac{1}{x}$$

$$\text{Solution } y \cdot I.F = \int O \cdot I.F$$

$$\frac{z}{x} = \int \frac{1}{x^2} dx + C$$

$$\frac{1}{xy} = \int \frac{1}{x^2} dx + C \Rightarrow \frac{x^{-3+1}}{-3+1}$$

$$\boxed{\frac{1}{xy} = -\frac{1}{2x^2} + C}$$

$$2) \frac{dy}{dx} + y \cot x = \cos x$$

$$\text{sol } I.F = e^{\int \cot x dx}$$

$$I.F = e^{\log(\sin x)} = \sin x$$

$$\sin xy = \int \cos x \sin x$$

$$y \sin u = \int \frac{1}{2} \cos 2u du \Rightarrow \int \frac{\sin 2u}{2}$$

$$\boxed{y \sin u = -\frac{\cos 2u}{2} + C} \quad \underline{\text{Ans}}$$

$$3) (x^2+1) \frac{dy}{dx} + 2xy = u^2$$

$$\text{sol } \frac{(x^2+1)}{(x^2+1)} \frac{dy}{dx} + \frac{2x}{(x^2+1)} y = \frac{u^2}{(x^2+1)}$$

$$\text{let } (x^2+1) = t \Rightarrow 2x = \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{t} = \frac{u^2}{t}$$

$$I.F = e^{\int \frac{1}{t} dt} = e^{\log t} \Rightarrow I.F = t$$

$$\Rightarrow t \cdot y = \int t \cdot \frac{u^2}{t} + C$$

$$\Rightarrow (1+x^2)y = \frac{x^3}{3} + C \quad \underline{\text{Ans}}$$

$$\Rightarrow \frac{\partial M}{\partial y} = 6x^2y - 6xy^2$$

$$\Rightarrow \frac{\partial N}{\partial x} = 6x^2y - 6xy^2$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\text{Now } \int M dx + \int N dy \text{ (terms not containing } x \text{)}$$

$$= \int (5x^4 + 3x^2y^2 - 2xy^3) dx + \int -5y^4 dy = C$$

$$= \frac{5x^5}{5} + \frac{3x^3y^2}{3} - \frac{2x^2y^3}{2} - \frac{5y^5}{5} = C$$

$$\boxed{x^5 + x^3y^2 - x^2y^3 - y^5 = C} \quad \underline{\text{Ans}}$$

Equation Reducible to Exact :-

Case I: If the given eqⁿ

$Mdx + Ndy = 0$ is homogeneous and $Mx + Ny \neq 0$, then.

$\frac{1}{Mx+Ny}$ is an I.F $\int \frac{dy}{dx} = \frac{x^3+y^3}{xy^2}$

$$\Rightarrow (x^3+y^3)dx - (xy^3)dy = 0$$

$$M = (x^3+y^3)$$

$$N = -(xy^3)$$

$$\frac{\partial M}{\partial y} = 3y^2 \quad \frac{\partial N}{\partial x} = -3xy^2$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\text{I.F} = \frac{1}{Mx+Ny} = \frac{1}{(x^3+y^3)x + (-xy^3)y}$$

Multiply I.F in given eqⁿ:

$$\frac{(x^3+y^3)dx}{(x^3+y^3)x - (xy^3)y} - \frac{xy^3 dy}{(x^3+y^3)x - (xy^3)y} = 0$$

$$\Rightarrow \frac{(x^3+y^3)dx}{x^4} - \frac{xy^3 dy}{x^4} = 0$$

$$\Rightarrow \left(\frac{x^3+y^3}{x^4} \right) dx - \frac{y^3}{x^3} dy = 0$$

$$\frac{\partial M}{\partial y} = 0 + 3y^2 \frac{1}{x^4}$$

$$\frac{\partial N}{\partial x} = \frac{3y^3}{x^4}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

∴ solution is:-

$$\int Mdx + \int Ndy = C \quad (N \text{ contains } x)$$

$$\Rightarrow \int \left(\frac{x^3}{x^4} + \frac{y^3}{x^4} \right) dx = \int \frac{y^3}{x^3} \frac{dy}{x}$$

$$\Rightarrow \log x + y^3 \left(\frac{-1}{3x^3} \right) = C$$

$$\boxed{\log x - \frac{y^3}{3x^3} = C}$$

Case II: - If eqⁿ is of the form

$$f_1(x,y)ydx + f_2(x,y)x dy = 0$$

then, I.F = $\frac{1}{Mx-Ny}$, $Mx-Ny \neq 0$

$$\Rightarrow y(xy + 2x^2y^2)dx + x(y - x^2y^2)dy = 0$$

$$\text{Here, } M = xy^2 + 2x^2y^3$$

$$N = x^2y - x^3y^2$$

$$\frac{\partial M}{\partial y} = 2xy + 6x^2y^2$$

$$\frac{\partial N}{\partial x} = 2xy - 3x^2y^2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$I.f = \frac{1}{Mx - Ny}$$

$$= \frac{1}{(xy^2 + 2x^2y^3)x - (x^2y - x^3y^2)y}$$

$$= \frac{1}{x^2y^2 + 2x^3y^3 - x^2y^2 + x^3y^3}$$

$$I.f = \frac{1}{3x^3y^3}$$

Multiplying I.f in eq

$$\frac{xy^2 + 2x^2y^3}{3x^3y^3} dx + \frac{x^2y - x^3y^2}{3x^3y^3} dy = 0$$

$$\Rightarrow \left(\frac{dx}{3x^2} + \frac{2}{3} \frac{y}{x} \right) dx + \left(\frac{1}{3xy} - \frac{1}{3y} \right) dy = 0$$

$$\int M dx + \int N dy (x \text{ const}) = C$$

$$\int \left(\frac{1}{3x^2} + \frac{2}{3} \frac{y}{x} \right) dx + \int -\frac{1}{3y} dy = C$$

$$+ \left[-\frac{1}{3xy} + \frac{2}{3} y \log x + -\frac{1}{3} \log y \right] = C$$

$$\Rightarrow \boxed{2 \log x - \frac{1}{xy} - \log y = C}$$

Q H.W

$$(1) (y - 2y^2) dx - (1 + x^2y) dy = 0$$

$$(2) (e^y + 1) \cos x dx + e^y \sin x dy = 0$$

$$(e^y + 1) \sin x = C$$

Rule III :-

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \text{function of } x \text{ only}$$

$$I.F = e^{\int f(x) dx}$$

⇒ Rule IV :-

$$\frac{1}{M} \left(\frac{\partial M}{\partial x} - \frac{\partial M}{\partial y} \right) = \text{function of } y \text{ only}$$

$$I.F = e^{\int f(y) dy}$$

$$\Rightarrow (2x \log x - xy) dy + 2y dx = 0$$

$$\Rightarrow 2y dx + (2x \log x - xy) dy = 0 \rightarrow (1)$$

$$\frac{\partial M}{\partial y} = \frac{d}{dy} 2y \Rightarrow 2 (y \text{ const.})$$

$$\Rightarrow \frac{\partial N}{\partial x} = 2x \cdot \frac{1}{x} + 2 \log x - y$$

$$\text{Since: } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Equation is not exact.

Now,

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$$

$$\frac{1}{2x \log x - xy} (2 - 2 - 2 \log x + y)$$

$$\frac{1}{(2x \log x - xy)} (y - 2 \log x)$$

$$\Rightarrow x \left(\frac{y - 2 \log x}{2 \log x - y} \right)$$

$= -\frac{1}{x}$ This is a function of x

$$I.F. = e^{\int \frac{1}{x} dx} = e^{\int \frac{1}{x} dx} = \frac{1}{x}$$

multiply eq (1) by $\frac{1}{x}$

$$\Rightarrow \frac{2y}{x} dx + (2 \log x - xy) dy = 0$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{2}{x} \quad \& \quad \frac{\partial N}{\partial x} = \frac{2}{x}$$

Solution of given eqn is

$$\int M dx + \int N dy \quad (\text{not } x)$$

$$= \int \frac{2y}{x} dx + \int -y dy$$

$$Ans = \boxed{2y \log x - \frac{y^2}{2} = C}$$

(4) case ②

$$Soln: (xy^3 + y) dx + 2(x^2y^2 + x + y^4) dy = 0$$

$$Sol \quad \frac{\partial M}{\partial y} = 3xy^2 + 1 \quad \frac{\partial N}{\partial x} = 4xy^2 + 2$$

Not equal.

$$I.F. = \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$\Rightarrow \frac{1}{(xy^3 + y)} (4xy^2 + 2 - 3xy^2 - 1)$$

$$= \frac{1}{(xy^3 + y)} (xy^2 + 1)$$

$$\Rightarrow \boxed{\frac{1}{y}}$$

$$I.F. = e^{\int \frac{1}{y} dy} = \boxed{y}$$

(multiply by y in ①)

$$(xy^4 + y^2) dx + (2x^2y^3 + x + 2y^5) dy$$

$$\frac{\partial M}{\partial y} = 4xy^3 + 2y$$

$$\frac{\partial N}{\partial x} = 4xy^3 + 2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution

$$\int M dx + \int N dy = C$$

$$\int (4xy^3 + 2y) dx + \int (4xy^3 + 2y^5) dy$$

$$\int (xy^4 + y^2) dx + \int 2x^2y^3 + 2xy + 2y^5 dy$$

$$\Rightarrow \int \frac{2^2 y^4}{2} + \frac{2}{4} x^2 y^4 + xy^2 + \frac{2y^6}{6}$$

$$\Rightarrow \frac{x^2 y^4}{2} + \frac{x^2 y^4}{2} + xy^2 + \frac{y^6}{3}$$

$$\Rightarrow \boxed{\frac{x^2 y^4}{2} + \frac{2xy^2}{2} + \frac{y^6}{3} = C}$$

$$\Rightarrow \boxed{\frac{x^2 y^4}{2} + xy^2 + \frac{y^6}{3} = C}$$

H.W

$$1) (y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$$

$$Ans: xy + \frac{2x}{y^2} + y^2 = C$$

$$2) y + \left(\frac{y^3}{3} + \frac{x^2}{2}\right) dx + \left(\frac{1+y^2}{4}\right) x dy = 0$$

$$Ans: 3yx^4 + y^3x^4 + x^6 = 12C = Ans$$

$$\frac{\partial}{\partial y} (y^4 + 2y) \Rightarrow 4y^3 + 2$$

$$\frac{\partial}{\partial x} (xy^3 + 2y^4 - 4x) \Rightarrow y^3 - 4$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\text{multiply } \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$\Rightarrow \frac{1}{(y^4 + 2y)} (y^3 - 4 - 4y^3 - 2)$$

$$= \frac{1}{y^4 + 2y} (-3y^3 - 6)$$

$$\Rightarrow \frac{-3(y^3 + 2)}{y(y^3 + 2)} \Rightarrow -\frac{3}{y}$$

this is a function of y

$$\therefore I.F = e^{\int -\frac{3}{y} dy}$$

$$= e^{-3 \log y}$$

$$\boxed{I.F = \frac{1}{y^3}}$$

multiply by $\frac{1}{y^3}$ in Q

$$\frac{(y^4 + 2y)}{y^3} dx + \frac{(xy^3 + 2y^4 - 4x)}{y^3} dy = 0$$

$$\left(\frac{y^3 + 2}{y^2}\right) dx +$$

$$\left(y + \frac{2}{y^2}\right) dx + \left(x + 2y - \frac{4x}{y^3}\right) dy$$

$$\frac{\partial M}{\partial y} = 1 - \frac{4}{y^3}$$

$$\frac{\partial N}{\partial x} = 1 - \frac{4}{y^3}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\therefore \int M dx + \int N dy = 0$$

(y con) (not containing x)

$$\int \left(y + \frac{2}{y^2}\right) dx + \int (2y) dy = C$$

$$\Rightarrow \boxed{xy + \frac{2x}{y^2} + y^2 = C}$$

$$2) y + \left(\frac{y^3}{3} + \frac{x^2}{2}\right) dx + \left(\frac{1+y^2}{4}\right) x dy = 0$$

$$Ans: \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(y + \frac{y^3}{3} + \frac{x^2}{2}\right)$$

$$\Rightarrow 1 + y^2 + 0$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} \left(\frac{x}{4} + \frac{xy^2}{4}\right) = \frac{1}{4} + \frac{y^2}{4}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\therefore \text{multiply by } \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$$

$$= \frac{4}{(x+xy^2)} \left(1+y^2 - \frac{1}{4} - \frac{y^2}{4} \right) \rightarrow (1)$$

$$= \frac{4}{x(1+y^2)} \left(\frac{3}{4} + \frac{3y^2}{4} \right)$$

$$= \frac{4x^3}{4x(1+y^2)} (1+y^2)$$

$$= \frac{3}{x}$$

$$\text{I.F.} = e^{\int \frac{3}{x} dx} = e^{3 \log x} = x^3$$

multiply x^3 in in (1)

$$= \left(yx^3 + \frac{11x^3y^3}{3} + \frac{11x^6}{2} \right) dx +$$

$$\text{Solution} \left(\frac{x^4}{4} + \frac{46x^2y^2}{4} \right) dy = 0$$

$$\int M dx + \int N dy = 0$$

$$= \int \left(yx^3 + \frac{11x^3y^3}{3} + \frac{11x^6}{2} \right) dx +$$

$$\int 0 dx = C$$

$$= \frac{yx^4}{4} + \frac{x^4y^3}{12} + \frac{x^6}{12} = C$$

multiply by 12

$$R_1 = 3yx^4 + y^3x^4 + x^6 = 12C$$

Case 5 :- If the given differential eqⁿ $Mdx + Ndy = 0$

having the form

$$x^a y^b (my dx + ndy) + x^p y^q (py dx + qx dy) = 0$$

where a, b, m, n, p, q are constant then it is

I.F $\Rightarrow x^h y^k$ then given diff. eqⁿ after multiplying $x^h y^k$ must become exact

$$\therefore \left[\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \right]$$

then we shall get the value of h, k thus the eqⁿ will be solved.

$$(y^3 - 2x^2y) dx + (2xy^2 - x^3) dy = 0 \rightarrow (1)$$

$$\frac{\partial M}{\partial y} = 3y^2 - 2x^2$$

$$\frac{\partial N}{\partial x} = 2y^2 - 3x^2$$

$$\text{Since } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ eqⁿ (1) is not exact

$$\rightarrow y^3 dx - 2x^2 y dx + 2xy^2 dy - x^3 dy$$

$$\rightarrow y^3 dx + 2xy^2 dy - 2x^2 y dx - x^3 dy$$

$$= y^2 (y dx + 2x dy) - x^2 (2y dx + x dy) = 0$$

let $I F = x^h y^k \rightarrow$ multiply by $\circ$

$$x^h y^k (y dx + x dy) - x^h y^k (2y dx + x dy) = 0$$

$$\Rightarrow x^h y^k (y^3 - 2x^2 y) dx + x^h y^k (2xy^2 - x^3) dy = 0$$

$$\Rightarrow \left(x^h y^{k+3} - 2x^{h+2} y^{k+1} \right) dx + \left(2x^{h+1} y^{2+k} - x^3 y^k \right) dy = 0$$

$$\frac{\partial M}{\partial y} = (k+3)x^h y^{k+2} - (k+1)2x^{h+2} y^k$$

$$\frac{\partial N}{\partial x} = 2(h+1)x^h y^{2+k} - (3+h)x^{h+2} y^k$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$(k+3)x^h y^{k+2} - 2(k+1)x^{h+2} y^k = 2(h+1)x^h y^{k+2} - (h+3)x^{h+2} y^k$$

$$\Rightarrow k+3 = 2(h+1) \rightarrow (eq)$$

$$\Rightarrow 2(k+1) = h+3$$

$$\Rightarrow k+3 = 2(h+1) \& 2k+1 = h+3$$

$$\boxed{k=1 \& h=1} \text{ Ans}$$

put value of k & h in eq

$$\Rightarrow (xy^4 - 2x^3 y^2) dx + (2x^2 y^3 - x^4 y) dy = 0$$

Soln

$$\int M dx + \int N dy = C$$

$$(xy^4 - 2x^3 y^2) dx + \int dy = C$$

$$\text{Ans. } \boxed{-\frac{x^2 y^4}{2} - \frac{x^4 y^2}{2} = C}$$

$$\text{Q. 11. } (2y dx + 3x dy) + 2xy(3y dx + 4 dy)$$

$$\text{Ans. } x^2 y^3 (1 + 2xy) = C$$

Linear Diff. Eqⁿ with constant coefficient

An eqⁿ of the form

$$\frac{dy}{dx} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + a_2 \frac{d^{n-2}y}{dx^{n-2}} + \dots + a_n y = 0$$

where,

$$a_1, a_2, a_3, \dots, a_n = \text{constant} \&$$

$\circ$ is a function of x is called linear diff. eqⁿ with constant coefficient.

$\circ$ If we denote

$$\frac{d}{dx}, \frac{d^2}{dx^2}, \frac{d^3}{dx^3}, \dots, \frac{d^n}{dx^n} \text{ by the operator } D, D^2, D^3, \dots, D^n$$

then eqⁿ become

$$D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \dots + a_{n-1} D y + a_n y = 0$$

$$(D^n + D^{n-1} + D^{n-2} + \dots + D + 1)y = 0 \quad (1)$$

$$[F(D)]y = 0 \rightarrow (2)$$

by solving L.H.S. & complementary function (C.F)

by solving R.H.S. = particular Integral (P.I)

Solution of given eqⁿ =

$$C.F + P.I = y$$

The complete solution of eqⁿ - (2)

$$y = C.F + P.I$$

methods for finding C.F.

1. In finding the comp C.F.

R.H.S of given eqⁿ is replaced by 0.

$$\text{i.e. } [F(D)]y = 0$$

2. Write the auxiliary eqⁿ replaced 0 by m

$$[F(m)]y = 0$$

Q1 Solve:

$$(2) \frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 0$$

$$\text{Sol } \frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 0$$

$$D^2y + 5Dy + 4y = 0$$

$$(D^2 + 5D + 4)y = 0$$

make auxiliary eqⁿ (y ko - det)

$$[m^2 + 5m + 4]y = 0$$

$$(m^2 + 4m + m + 4) = 0$$

$$m = -4 \text{ \& } m = -1$$

Case: 1:- When roots are real & distinct then diff. eqⁿ having C.F

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$$

Solution:-

$$\text{Ans } \Rightarrow y = C_1 e^{-4x} + C_2 e^{-x}$$

Q2 Case: 2:- When roots are real & equal

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$$

$$D^2y - 6Dy + 9y = 0$$

$$(D^2 - 6D + 9)y = 0$$

Its auxiliary eqⁿ is
 $(m^2 - 6m + 9) = 0$
 $m = 3, 3$

Roots are real & equal.

The solⁿ of given eqⁿ is

if it has 2 roots equal =
 $m_1 = m_2, m_3, m_4, m_5, \dots$

Solution $\Rightarrow$ $y = (C_1 + C_2 x)e^{m_1 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$

if 3 roots are equal =
 $m_1 = m_2 = m_3 = \dots$

solⁿ $\Rightarrow y = (C_1 + C_2 x + C_3 x^2)e^{m_1 x} + C_4 x^{m_4} + C_5 x^{m_5}$

$\Rightarrow$ The general solution of (1) is

$y = (C_1 + C_2 x)e^{3x}$

if roots are imaginary

$m_1 = a + ib, m_2 = a - ib$

$y = e^{ax} (C_1 \cos bx + C_2 \sin bx)$

$(D^4 - 1)y = 0$

$f(D)y = 0$ (making Auxiliary Equation)

$m^4 - 1 = 0$

$(m^2 - 1)(m^2 + 1) = 0$

$(m - 1)(m + 1)(m^2 + 1) = 0$

$m = 1, -1, \pm i$

Solution for real root

$y = C_1 e^x + C_2 e^{-x}$

Solution for imaginary root

$a + i$ & $a - i$

Solⁿ

$y = e^{ax} (C_1 \cos bx + C_2 \sin bx)$

$y = e^{ax} (C_1 \cos x + C_2 \sin x)$

$y = C_1 \cos x + C_2 \sin x$

final solⁿ is

$y = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x$

if roots are equal & imaginary

$m_1 = m_2 = a + ib$

$m_3 = m_4 = a - ib$

$y = e^{ax} [C_1 + x C_2] \cos bx + (C_3 + x C_4) \sin bx$

$(D^4 + 8D^2 + 16)y = 0$

$m^4 + 8m^2 + 16 = 0$

$m^4 + 4m^2 + 4m^2 + 16$

$m^2(m^2 + 4) + 4(m^2 + 4)$

$(m^2 + 4)(m^2 + 4) =$

$\Rightarrow m = \pm i 2, m = \pm i 2$

$m_1 = m_2 = a + ib$

$m_3 = m_4 = a - ib$

Solⁿ, $y = (C_1 + x C_2) \cos 2x + (C_3 + x C_4) \sin 2x$

$$y = e^{ax} [(c_1 + xc_2) \cos 2x + (c_3 + xc_4) \sin 2x]$$

$$y = e^{ax} [(c_1 + xc_2) \cos 2x + (c_3 + xc_4) \sin 2x]$$

vi) when roots are imaginary
 $m = \alpha \pm \sqrt{\beta}$

$$\text{Sol}^n \Rightarrow \text{C.F.} = e^{\alpha x} (c_1 \cosh \sqrt{\beta} x + c_2 \sinh \sqrt{\beta} x)$$

vii) when roots of A.E are same
 limited & distinct

$$m = m_1 = m_2 = m_3 = m_4, m_4, m_5$$

$$\text{Sol}^n = \text{C.F.} (c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_2 x} + c_5 e^{m_3 x}$$

$$\text{Q:- } \frac{d^5 y}{dx^5} - \frac{d^3 y}{dx^3} = 0$$

$$\Rightarrow (D^5 - D^3) y = 0$$

$$\text{Auxiliary eq}^n$$

$$m^5 - m^3 = 0$$

$$m^3 (m^2 - 1) = 0$$

$$m = 0, 0, 0, 1, -1$$

$$\text{Sol}^n \Rightarrow (c_1 + c_2 x + c_3 x^2) e^{0x} + (c_4 + c_5 x) e^x + c_6 e^{-x}$$

Rules to find Particular Integral (P.I)

$$f(D)y = 0$$

$$P.I = \frac{0}{f(D)}$$

Case I: when $\phi = e^{ax}$

$$P.I = \frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)}$$

replace D by a when $f(a) \neq 0$

when $f(a) = 0$

$$P.I = \frac{x e^{ax}}{f'(D)} \Rightarrow \frac{x e^{ax}}{f'(a)}$$

$f'(a) \neq 0$

when $f'(a) = 0$

$$P.I = \frac{x^2 e^{ax}}{f''(D)} \Rightarrow \frac{x^2 e^{ax}}{f''(a)}$$

$f''(a) \neq 0$

$$\text{Q} \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 9e^{3x}$$

Solⁿ C.F

$$f(D^2 + 6D + 9)y = \text{C.F}$$

Auxiliary eqⁿ

$$m^2 + 6m + 9 = 0$$

RHS = 0 for C.F

$$m^2 + 3m + 3m + 1 = 0$$

$$m = -3, -3$$

$$C.F. = (C_1 x + C_2 x) e^{-3x}$$

P.I. $F(D)y = 0$

$$P.I. = \frac{0}{F(D)}$$

$$y = \frac{5e^{3x}}{F(D^2 + 6D + 9)}$$

$y =$ Substitute D by 3

$$y = \frac{5e^{3x}}{F(9 + 6 \times 3 + 9)} = \frac{5e^{3x}}{F(18 + 18)}$$

$$= \frac{5e^{3x}}{F(36)} = \frac{5}{36} e^{3x}$$

Soln $\Rightarrow C.F. + P.I.$

$$y = (C_1 + C_2 x) e^{-3x} + \frac{5}{36} e^{3x}$$

$$(D^2 + D - 2)y = e^x$$

$$(m^2 + m - 2)$$

$$m = -2, 1$$

$$C.F. = C_1 e^{-2x} + C_2 e^x$$

$$P.I. = \frac{e^x}{D^2 + D - 2} = \frac{e^x}{1 + 1 - 2} = \frac{e^x}{0}$$

differentiate it

$$= \frac{x e^x}{2D + 1} \Rightarrow \frac{x e^x}{2 + 1} = \frac{x e^x}{3}$$

Soln $\Rightarrow y = C.F. + P.I.$

$$y = C_1 e^x + C_2 e^{-2x} + \frac{x e^x}{3}$$

Case II: $[f(D)y = 0]$

$\nexists 0 = \sin ax$ or $\cos ax$

then:

$$P.I. = \frac{1}{f(D^2)} \sin ax \Rightarrow \frac{1}{f(-a^2)} \sin ax$$

For $f(-a^2) \neq 0$

$\nexists f(-a^2) = 0$ then $P.I. = \frac{x \sin ax}{f'(D^2)}$

$$\Rightarrow \frac{x \sin ax}{f'(-a^2)}$$

$$(D^2 + 4)y = \sin 3x$$

Sol $(D^2 + 4)y = \sin 3x$

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

$$C.F. = e^{0x} (C_1 + x C_2)$$

$$= e^{0x} (C_1 \cos 2x + C_2 \sin 2x)$$

$$P.I. = \frac{0}{f(D)} = \frac{\sin 3x}{(D^2 + 4)}$$

$$\Rightarrow \frac{\sin 3x}{(-(3^2 + 4))} = \frac{\sin 3x}{-9 - 4} = \frac{\sin 3x}{-13}$$

$\nexists$ Complete soln $\Rightarrow y = C.F. + P.I.$

$$\Rightarrow C_1 \cos 2x + C_2 \sin 2x - \frac{1}{13} \sin 3x$$

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \cos 2x$$

$$\Rightarrow (D^2 + D + 1)y = \cos 2x$$

$$D^2 + D + 1 = 0$$

$$m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$C.F. = e^{-\frac{x}{2}} \left(C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right)$$

$$P.I. = \frac{\cos 2x}{(-D^2 + D + 1)} = \frac{\cos 2x}{(-(2)^2 + D + 1)}$$

$$= \frac{\cos 2x}{(D-3)} \cdot \frac{(D+3)}{(D+3)}$$

$$= \frac{(D+3)(\cos 2x)}{D^2 - 9}$$

$$\Rightarrow \frac{(D+3)(\cos 2x)}{-(2)^2 - 9} = \frac{(D+3)(\cos 2x)}{-13}$$

$$= \frac{D \cos 2x}{-13} + \frac{3 \cos 2x}{-13} \quad (\text{differentiate})$$

$$\Rightarrow \frac{-2 \sin 2x + 3 \cos 2x}{-13}$$

$$y = C.F. + P.I.$$

$$= e^{-\frac{x}{2}} \left(C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right) - \frac{1}{13} (-2 \sin 2x + 3 \cos 2x)$$

Q. 120 ①

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = \sinh x$$

Ans

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

It is

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = 4 \cos^2 x$$

$$| \cos^2 x = 2 \cos^2 x - 1 |$$

$$\left(\frac{1}{2} \right) = \frac{b}{2} e^{\frac{b}{2} x}$$

Case III :-

$$\text{Case } \frac{1}{f(x)} = [F(x)]^{-1} x^n$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$Q. (D^2 + 5D + 4)y = (3-2x)$$

$$3y \quad C.F. \Rightarrow m^2 + 5m + 4$$

$$(m^2 + 4m + m + 4)$$

$$(m+4)(m+1)$$

$$m = -4, -1$$

$$C.F. = C_1 e^{-4x} + C_2 e^{-x}$$

$$P.I. = \frac{3-2x}{(D^2 + 5D + 4)}$$

$$= \frac{1}{4 \left(\frac{D^2}{4} + \frac{5D}{4} + 1 \right)} (3-2x)$$

$$= \frac{1}{4 \left(\frac{D^2}{4} + \frac{5D}{4} + 1 \right)} (3-2x)$$

$$\begin{aligned} & \Rightarrow \frac{1}{4} \left(\frac{D^2}{4} + \frac{5D}{4} + 1 \right)^{-1} (3-2u) \\ & = \frac{1}{4} \left(1 + \left(\frac{D^2}{4} + \frac{5D}{4} \right) \right)^{-1} (3-2u) \\ & \Rightarrow \frac{1}{4} \left[1 - \left(\frac{D^2}{4} + \frac{5D}{4} \right) + \left(\frac{D^2}{4} + \frac{5D}{4} \right)^2 \right. \\ & \quad \left. \dots \right] (3-2u) \end{aligned}$$

$$\begin{aligned} & \Rightarrow \frac{1}{4} \left[(3-2u) - \left(\frac{15D}{4} + \frac{D^2}{4} \right) (3-2u) + \dots \right] \\ & = \frac{1}{4} \left[3-2u - \frac{15}{4} D - \frac{3}{4} D^2 + \frac{100D}{4} + \frac{20D^2}{4} \right] \\ & = \frac{1}{4} \left[(3-2u) - \frac{5}{4} D (3-2u) \right] \\ & \Rightarrow \frac{1}{4} \left(3-2u - 0 + \frac{5}{2} \right) \end{aligned}$$

$$\Rightarrow \frac{1}{4} \left(\frac{11}{2} - 2u \right)$$

$$P.I. = \frac{1}{8} (11-4u)$$

Solution:- $y = C.F + P.I$

$$y = C_1 e^{-4x} + (2e^{-x} + \frac{1}{8} (11-4u))$$

Case: 4: $\frac{1}{f(D)} \phi(u) \Rightarrow e^{ax} \frac{1}{f(D+a)} \phi(u)$

$$\frac{\theta}{(D^2-4D+4)} = x^2 e^{2x}$$

$$C.F = (C_1 + C_2 x) e^{2x}$$

$$P.I = e^{2x} \frac{1}{((D+2)^2 - 4(D+2) + 4)} x^2$$

$$\Rightarrow e^{2x} \frac{1}{(D^2+4+2D-4D-8+4)} x^2$$

$$e^{2x} \frac{1}{(D^2-2D)} x^2 \quad \left[\frac{1}{D^2} = \text{Integration two times} \right]$$

$$\left[\frac{e^{2x}}{D} \frac{1}{3} \Rightarrow e^{2x} \frac{x^4}{12} \right]$$

$$Ans: y = (C_1 + C_2 x) e^{2x} + e^{2x} \frac{x^4}{12}$$

Case: 5 $D^2 - 2D + 4 = x \sin \theta$

$$P.I = \frac{1}{f(D)} x \phi(u)$$

$$P.I = \left[x \frac{1}{f(D)} - \frac{f'(D)}{f(D)^2} \right] \phi(u)$$

Case 6: when θ is any other function of x Reduce $f(D)$ into linear factor.

$$\text{Let } f(D) = (D-m_1)(D-m_2) \dots (D-m_n)$$

$$\text{then } P.I = \frac{1}{f(D)} \theta$$

$$= \frac{\theta}{(D-m_1)(D-m_2) \dots (D-m_n)}$$

$$(D-m_1)(D-m_2) \dots (D-m_n)$$

$$= \frac{A_1}{(D-m_1)} \theta + \frac{A_2}{(D-m_2)} \theta \dots$$

$$\textcircled{1} \frac{1}{(D-\alpha)} \theta = e^{\alpha x} \int e^{-\alpha x} \theta dx$$

$$\textcircled{2} \frac{1}{(D+\alpha)} \theta = e^{-\alpha x} \int e^{\alpha x} \theta dx$$

$$e^{i\theta} = \cos \theta + i \sin \theta \rightarrow (i)$$

$$e^{-i\theta} = \cos \theta - i \sin \theta \rightarrow (ii)$$

$$y \frac{dy}{dx} - 2y \frac{dy}{dx} + 2y = e^x \tan x$$

$$-2y \frac{dy}{dx} = e^x (C_1 \cos x + C_2 \sin x)$$

$$P.I = \frac{1}{D^2 - 2D} e^x \tan x$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} \tan x$$

$$= e^x \frac{1}{D^2 + 1} \tan x$$

$$= e^x \frac{1}{(D+i)(D-i)} \tan x$$

$$\frac{e^x}{2i} \left[\frac{1}{(D-i)} - \frac{1}{(D+i)} \right] \tan x - (i)$$

$$\frac{\tan x}{(D-i)} = e^{-ix} \int e^{ix} \tan x dx$$

$$\Rightarrow e^{-ix} \int (\cos x - i \sin x) \tan x dx$$

$$= e^{-ix} \int \left[\sin x - i \frac{\sin^2 x}{\cos x} \right] dx$$

$$\Rightarrow e^{-ix} \int \left[\sin x - i \left(\frac{1 - \cos^2 x}{\cos x} \right) \right] dx$$

$$\Rightarrow e^{-ix} \left[-\cos x - i \sec x + i \cos x \right] dx$$

Similarly changing $-i$ to i we have
 $e^{ix} [-\cos x + i \log(\sec x + \tan x) - i \sin x] + (3)$

$$\Rightarrow \frac{e^x}{2i} \int e^{ix} [-\cos x - i \log(\sec x + \tan x)]$$

$$+ i \sin x - e^{-ix} \cos x + \log(\sec x + \tan x) - i \sin x$$

$$\Rightarrow e^x [-\cos x \left(\frac{e^{ix} + e^{-ix}}{2i} \right) - \log(\sec x + \tan x) \left(\frac{e^{ix} + e^{-ix}}{2} \right) + \sin x \left(\frac{e^{ix} - e^{-ix}}{2} \right)]$$

$$\Rightarrow e^x [-\cos x \sin x - \cos x \log(\sec x + \tan x) + \sin x \cos x]$$

$$P.I \Rightarrow -e^x \cos x \log(\sec x + \tan x)$$

$$\boxed{y = C.F + P.I}$$

Linear diff. equation with variable coefficient

Cauchy's eqⁿ:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 x \frac{dy}{dx} + a_0 y = 0$$

put $x = e^z$

$\frac{d}{dz} = D$

$Z = \log x$

$x \frac{dy}{dx} = \frac{dy}{dz} \times \frac{dx}{dz}$

$x \frac{dy}{dx} = \frac{dy}{dz}$

$x \frac{dy}{dx} = D y$

$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$

$\Rightarrow \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dz} \right)$

$\Rightarrow -\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x} \frac{d^2 y}{dz^2} \frac{dz}{dx}$

$\Rightarrow -\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x} \frac{d^2 y}{dz^2} \times \frac{1}{x}$

$\Rightarrow \frac{1}{x^2} \left(-\frac{dy}{dz} + \frac{d^2 y}{dz^2} \right)$

$\frac{x^2 d^2 y}{dx^2} = (D^2 - D)y$

$\frac{x^2 d^2 y}{dx^2} = D(D-1)y$

$\frac{x^3 d^3 y}{dx^3} = D(D-1)(D-2)y$

$\frac{x^2 d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$

put $x = e^z$, $z = \log x$

$D = \frac{d}{dz}$

$\Rightarrow D(D-1)y - 2Dy - 4y = e^{4z}$

$(D^2 - D)y - 2Dy - 4y = e^{4z}$

$(D^2 - 3D - 4)y = e^{4z}$

making auxiliary equation

$(D^2 - 4D + D - 4)y = e^{4z}$

$(D(D-4) + 1(D-4))$

$(m = 4, -1)$

C.F. $C_1 e^{4z} + C_2 e^{-z}$

P.I. $= \frac{e^{4z}}{D^2 - 3D - 4}$

$= \frac{e^{4z}}{16 - 12 - 4}$

$\Rightarrow \frac{e^{4z}}{0} \Rightarrow \frac{z e^{4z}}{2D - 3} = \frac{z e^{4z}}{5}$

P.I. $\Rightarrow \frac{z e^{4z}}{5}$

Solution $\Rightarrow$ C.F. + P.I.

$\Rightarrow C_1 e^{4z} + C_2 e^{-z} + \frac{z e^{4z}}{5}$

put $z = \log x$

$$y = c_1 x^4 + c_2 \frac{1}{x} + \frac{\log x}{5} x^4$$

$$Q \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \sin(\log x^2)$$

Sol put $x = e^z \quad z = \log x$

$$D(D-1)y + Dy + y = \sin(\log e^{2z})$$

$$(D^2 - D + D + 1)y = (\sin 2z)$$

$$(D^2 + 1)y = \sin 2z$$

$$D = \pm i$$

$$C.F = e^{0z} (c_1 \cos z + c_2 \sin z)$$

$$P.I = \frac{\sin 2z}{D^2 + 1} \Rightarrow \frac{\sin 2z}{-(2)^2 + 1}$$

$$P.I \Rightarrow \frac{\sin 2z}{-3}$$

$$\text{Solution } y = c_1 \cos z + c_2 \sin z - \frac{\sin 2z}{3}$$

put $z = \log x$

$$y = c_1 \cos \log x + c_2 \sin \log x - \frac{\sin 2 \log x}{3}$$

Legendre's diff. eqⁿ

A diff. eqⁿ of the form $(a+bx)^n D^n y + (a+bx)^{n-1} D^{n-1} y + \dots + a_n y = X \rightarrow (i)$

$$\text{put } [a+bx = e^z]$$

where $a, a_1, a_2, a_3, \dots$ are constants

where X is funⁿ of x is called Legendre's diff. eqⁿ.

$$z = \log(a+bx)$$

$$(a+bx) \frac{dy}{dx} = D y$$

$\Rightarrow$ substitute it :-

$$(a+bx)^2 \frac{d^2 y}{dx^2} = b^2 D(D-1)y$$

$$(a+bx)^3 \frac{d^3 y}{dx^3} = b^3 D(D-1)(D-2)y$$

$$\text{Ques:- } (1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin 2 \log(1+x)$$

$$\text{Sol } [1+x = a+bx = e^z]$$

$$z = \log(1+x)$$

$$D = \frac{d}{dz}$$

$$(D(D-1)y + Dy + y = \sin 2 \log x)$$

$$\Rightarrow D^2 y - Dy + Dy + y = \sin 2z$$

$$\Rightarrow D^2 y + y = \sin 2z$$

$$\Rightarrow (D^2 + 1)y = \sin 2z$$

$$C.F = C_1 \cos z + C_2 \sin z$$

$$P.I = \frac{\sin 2z}{(D^2+1)} = \frac{\sin 2z}{-4+1} = \frac{\sin 2z}{-3}$$

$$\text{Solution} = C.F + P.I$$

$$y = C_1 \cos z + C_2 \sin z - \frac{\sin 2z}{3}$$

$$y = C_1 \cos \log(1+x) + C_2 \sin \log(1+x) - \frac{1}{3} \sin 2 \log(1+x)$$

Q. 10

$$(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \{ \log(1+x) \}$$

$$\text{Sol. let } 1+x = e^z \Rightarrow z = \log(1+x)$$

$$\Rightarrow (D(D-1) + D + 1)y = 4 \cos z$$

$$\Rightarrow (D^2 - D + D + 1)y = 4 \cos z$$

$$\Rightarrow (D^2 + 1)z = 4 \cos z$$

$$C.F \Rightarrow (D^2 + 1)z$$

Convert it into auxiliary eqn

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i$$

$$C.F = e^{a_1 x} (C_1 \cos b_1 x + C_2 \sin b_1 x)$$

$$= e^{0x} (C_1 \cos x + C_2 \sin x)$$

$$C.F = C_1 \cos z + C_2 \sin z$$

$$P.I = \frac{4 \cos z}{f(D)}$$

$$P.I = \frac{4 \cos z}{(D^2 + 1)} \Rightarrow \frac{4 \cos z}{-1 + 1} = \frac{4 \cos z}{0}$$

differentiate M.

$$P.I = \frac{4z \cos z \times D}{2D \times D}$$

$$P.I = \frac{4z \cos z \times D}{2D^2}$$

$$P.I = \frac{4z D \cos z}{2(-1)^2}$$

$$P.I = -\frac{4}{2} z (-\sin z)$$

$$P.I = 2z \sin z$$

$$\text{Soln } y = C.F + P.I$$

$$y = C_1 \cos z + C_2 \sin z + 2z \sin z$$

$$y = C_1 \cos \log(1+x) + C_2 \sin \log(1+x) + 2 \log(1+x) \sin \log(1+x)$$

$$2(3x+2)^2 \frac{d^2 y}{dx^2} + 3(3x+2) \frac{dy}{dx} - 3y = 3x^2 + 4x + 1$$

Method of variation of parameter

It enables to find the general solⁿ of any linear differential eqⁿ of second order when provided its C.F. given that P.I. of the differential eqⁿ is obtained by varying the parameter i.e. replacing the arbitrary constant in the C.F. by variable function

$$a^2 \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = x \text{ [constant] (i)}$$

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + Q(x)y = X \text{ [with variable co-efficient]}$$

$$\text{Now: C.F. } y_c = Ay_1 + By_2 \text{ --- (ii)}$$

In the method of variation of parameter, the arbitrary constants A, B in eqⁿ (ii) is replaced by two unknown funⁿ U & V . Thus, the P.I. of eqⁿ (i)

$$y_p = Uy_1 + Vy_2$$

Hence the required general solⁿ is!

$$y = y_c + y_p$$

to find the P.I. of $a^2 \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = x$

$$\text{like C.F. } \Rightarrow ay_1 + by_2$$

$$\text{then P.I. } \Rightarrow Uy_1 + Vy_2$$

$$\text{where } U = \int \frac{-y_2 x}{y_1 y_2' - y_1' y_2} dx$$

$$V = \int \frac{y_1 x}{y_1 y_2' - y_1' y_2} dx$$

$$\text{complete solⁿ :- } y = y_c + y_p$$

$$\frac{d^2 y}{dx^2} + y = \cos x$$

$$y(D^2 + 1) = \cos x$$

$$\text{auxiliary eqⁿ } = m^2 + 1 = 0$$

$$m = \pm i$$

$$\text{C.F. } y_c = A \cos x + B \sin x$$

$$\text{Hence: } y_1 = \cos x, y_2 = \sin x$$

$$\text{P.I.} = Uy_1 + Vy_2 \text{ where}$$

$$U = \int \frac{-y_2 x}{y_1 y_2' - y_1' y_2} dx$$

$$V = \int \frac{y_1 x}{y_1 y_2' - y_1' y_2} dx$$

$$U = \int \frac{-\sin x \cos x}{\cos^2 x + \sin^2 x} dx$$

$$= \int -dx = -x$$

$$V = \int \frac{\cos x \cos x}{1} dx$$

$$= -\int \cot x dx = \log \sin x$$

$$\therefore \text{P.I.} = -x \cos x + \sin x \log \sin x$$

General solⁿ :-

$$y = y_c + y_p$$

$$y = A \cos x + B \sin x - x \cos x + \sin x \log \sin x$$

Simultaneous Differential Eqⁿ:

If two or more independent variables are funⁿ of a single independent variable the eqⁿ involving their derivative are called simultaneous differential eqⁿ

Ex: $\frac{dx}{dt} + 4y = t$

$$\frac{dy}{dt} + 2x = e^t$$

The method of solving these eqⁿ is based on process of elimination as well as simultaneous algebraic equation.

Solve the eqⁿ:-

$$\frac{dx}{dt} = -wy, \frac{dy}{dt} = wx$$

on putting $\frac{d}{dt} = D$ in the above eqⁿ we have,

$$Dx + wy = 0 \rightarrow (i)$$

$$-wx + Dy = 0 \rightarrow (ii)$$

on multiplying eq(i) by w and eq(ii) by D and adding

$$Dwx + w^2y = 0$$

$$-Dwx + D^2y = 0$$

$$(D^2 + w^2)y = 0$$

$$m^2 + w^2 = 0$$

$$m = \pm iw$$

$$y = c_1 \cos wt + c_2 \sin wt$$

$$Dy = -c_1 w \sin wt + c_2 w \cos wt$$

put the value of $Dy = -c_1 w \sin wt + c_2 w \cos wt$ in eq(2)

put $-wx = c_1 w \sin wt + c_2 w \cos wt = 0$

we solve for x

$$D \frac{dx}{dt} + y = \sin t$$

$$\frac{dy}{dt} + x = \cos t$$

given that $x=2, y=0$

when $t=0$

Sol. let $\frac{d}{dt} = D$

$$Dx + y = \sin t \rightarrow (i)$$

$$Dy + x = \cos t \rightarrow (ii)$$

By elimination method multiply D in eq(ii)

$$D^2x + Dy = \sin t$$

$$Dx + D^2y = -\sin t$$

$$y - D^2y = \sin t + \sin t$$

$$y - D^2y = 2 \sin t$$

$$(1 - D^2)y = 2 \sin t$$

$$C.F = D^2 - 1 = 0$$

$$\Rightarrow D = \pm 1$$

$$C.F = c_1 e^x + c_2 e^{-x}$$

$$C.F = c_1 e^t + c_2 e^{-t}$$

$$P.I = \frac{1}{D^2 - 1} (-2 \sin t)$$

$$= \frac{-1}{-1} \sin t$$

$$\Rightarrow \sin t$$

$$y = c_1 e^t + c_2 e^{-t} + \sin t$$

$$\text{when } t=0, y=0$$

$$0 = c_1 e^0 + c_2 e^{-0} + \sin 0$$

$$0 = c_1 + c_2$$

$$\boxed{c_2 = -c_1}$$

$$y = c_1 e^t - c_1 e^{-t} + \sin t$$

$$y = c_1 \left(e^t - \frac{1}{e^t} \right) + \sin t$$

putting y in eq (ii)

$$x + Dy = \cos t \quad \rightarrow \text{differentiate it}$$

$$x + D(c_1 e^t - c_1 e^{-t} + \sin t) = \cos t$$

$$x + c_1 e^t + c_1 e^{-t} + \cos t = \cos t$$

$$\Rightarrow x + c_1 e^t + c_1 e^{-t} = 0$$

$$\text{when } t=0, x=2$$

$$\Rightarrow 2 + c_1 + c_2 = 0 \quad 2 = -c_1 - c_2$$

$$\Rightarrow c_1 + c_2 = -2 \quad 2 = -2c_1$$

$$c_1 + c_2 = 0 \quad \boxed{c_1 = -1}$$

$$\Rightarrow \boxed{c_1 = -1, c_2 = 1}$$

$$x = -c_1 e^t - c_2 e^{-t}$$

$$\boxed{x = e^t + e^{-t}} \quad \text{Ans}$$

$$y = c_1 e^t - c_1 e^{-t} + \sin t$$

$$\boxed{y = -e^t + e^{-t} + \sin t} \quad \text{Ans}$$

$$Q \frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 0$$

$$\frac{dy}{dt} + 5x + 3y = 0$$

$$\Rightarrow Dx + Dy + 2x + y = 0 \quad \text{--- (i)}$$

$$\Rightarrow Dy + 5x + 3y = 0 \quad \text{--- (ii)}$$

$$0x - 3x - 2y = 0$$

$$0x - 3x = 2y$$

$$(0-3)x = 2y$$

$$\Rightarrow 0-3 = 0$$

$$\boxed{D=3}$$

$$C.F. = \frac{2y}{D-3}$$

$$C.F. = \frac{2y}{1-3} = 2$$

$$S(D+1)x + S(D+1)y = 0$$

$$(D+3)(D+2)y + S(D+2)x = 0$$

$$S(D+1)y =$$

$$(D+3)(D+2)y - S(D+1)y = 0$$

$$y(D+3)(D+2) - S(D+1) = 0$$

$$y(D^2 + 2D + 3D + 6 - S - S) = 0$$

$$y(D^2 + 5D + 6 - 2S) = 0$$

$$y^2 \pm \dot{y}$$

$$C.F. = c_1 e^{st} + c_2 e^{-st}$$

$$P.F. = c_1 \cos t + c_2 \sin t$$

$$\boxed{y = c_1 \cos t + c_2 \sin t}$$

$$(D+2)x + (D+1)(c_1 \cos t + c_2 \sin t) = 0$$

$$Dx + 2x + (-c_1 \sin t) + c_1 \cos t + c_2 \cos t + c_2 \sin t = 0$$

$$\frac{Dx}{dt} + 2x - c_1 \sin t + c_1 \cos t + c_2 \cos t + c_2 \sin t = 0$$

$$(D^4 - D^3 - D^2 + 1)y = 0$$

$$D^2(D^2 - 1) - 1(D^2 - 1)$$

$$m^4(m^2 - 1) - 1(m^2 - 1) = 0$$

$$(m^2 - 1)(m^2 - 1)$$

$$Dy + 5x + 3y = 0$$

$$-c_1 \sin t + c_2 \cos t + 5x + 3(c_1 \cos t + c_2 \sin t) = 0$$

$$x = \frac{1}{5} (c_1 \sin t - c_2 \cos t - 3c_1 \cos t - 3c_2 \sin t)$$

1.10

$$\frac{d^2x}{dt^2} - 3x - 4y = 0$$

$$\frac{d^2y}{dt^2} + x + y = 0$$

$$\Rightarrow D^2x - 3x - 4y = 0 \Rightarrow (D^2 - 3)x - 4y = 0$$

$$D^2y + x + y = 0 \Rightarrow (D^2 + 1)y + x = 0$$

$\Rightarrow$ multiply $(D^2 - 3)$ in eq (ii)

$$m (D^2 - 3)x - 4y = 0 \quad (13)$$

$$(D^2 + 1)(D^2 - 3)y + (D^2 - 3)x = 0 \quad (14)$$

(13 - 14)

$$\Rightarrow (D^2 + 1)(D^2 - 3)y + (D^2 - 3)x - (D^2 - 3)x + 4y = 0$$

$$\Rightarrow (D^4 - 3D^2 + D^2 - 3)y + 4y = 0$$

$$\Rightarrow ((D^4 - 2D^2 - 3) + 4)y = 0$$

$$(D^4 - 2D^2 + 1)y = 0$$

Application of differential eqn

1. Electrical circuit :-

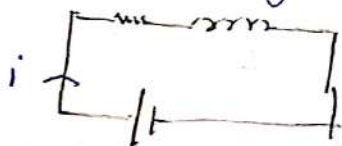
$$\rightarrow i = \frac{dq}{dt} \rightarrow \text{voltage drop across}$$

$$\rightarrow V = Ri \rightarrow R = \frac{V}{i}$$

$$\rightarrow L = L \frac{di}{dt}$$

$$\rightarrow C = q/c$$

L-R circuit :- let i be a current flowing in the circuit containing resistance R & an inductance L in series with voltage source E at any time t .



By voltage law:

$$Ri + L \frac{di}{dt} = E$$

$$\frac{Ri}{L} + \frac{1}{L} \frac{di}{dt} = \frac{E}{L}$$

$$\boxed{\frac{Ri}{L} + \frac{di}{dt} = \frac{E}{L}}$$

Linear eqn $\frac{dy}{dx} + Py = Q$

$$I.F. = e^{\int P dx}$$

$$\boxed{I.F. = e^{\frac{Rt}{L}}}$$

$$\text{Soln} \Rightarrow e^{\frac{Rt}{L}} i = \int \frac{E}{L} e^{\frac{Rt}{L}}$$

$$ie^{\frac{Rt}{L}} = \frac{E}{L} e^{\frac{Rt}{L}} \cdot \frac{L}{R} + C$$

$$\boxed{i = \frac{E}{R} + Ae^{-\frac{Rt}{L}}}$$

$$\Rightarrow \text{at } t=0, i=0, \Rightarrow A = -E/R$$

$$\boxed{i = \frac{E}{R} [1 - e^{-\frac{Rt}{L}}]}$$

C-R circuit :-

$$\boxed{R \frac{dq}{dt} + q/C = E}$$

let i be current in the circuit containing resistance R , L & capacitance C in series with voltage E and

an inductance of a $2H$ & R of 20Ω are connected in series with an emf E if the current is zero, when $t=0$, find current at the end of 0.01 , if $E = 100V$

$$\Rightarrow i = \frac{E}{R} [1 - e^{-\frac{Rt}{L}}]$$

$$\text{when } t=0, i=0$$

$$0 = \frac{100}{20} [1 - e^0]$$

$$\Rightarrow i = \frac{100}{20} [1 - e^{-\frac{20 \times 10^{-2}}{2}}]$$

$$\Rightarrow i = 5 [1 - e^{-1}]$$

$$i = 5 []$$

$$(i = 0.475 A \text{ approx})$$

Q Solve the equation $L \frac{dv}{dt} + Rv = E \sin \omega t$ then
 where L, R, E, ω are constant & discuss
 where $t \rightarrow \infty$.

$$F = ma$$

$$F = m \frac{d^2x}{dt^2}$$

$$F = m \frac{dv}{dt}$$

Q A moving body is opposed
 by F per unit mass of value
 $cx + b$ where x is distance per
 unit mass of value bx^2
 where x & v are displacement &
 velocity of the particle. Find
 the velocity of particles in terms of
 x if it starts from rest

$$F = m \frac{dv}{dt}$$

$$-v \frac{dv}{dx} = -cx - bx^2$$

$$\int v \frac{dv}{dx} + bx^2 = -cx \rightarrow (2)$$

$$\text{putting } v^2 = z$$

$$2v \frac{dv}{dx} = \frac{dz}{dx}$$

$$z \frac{dz}{dx} + bx^2 = -cx$$

$$z \frac{dz}{dx} + bx^2 = -cx$$

$$z \frac{dz}{dx} + bx^2 = -cx$$

$$z \frac{dz}{dx} + bx^2 = -cx$$

$$\int z \frac{dz}{dx} + bx^2 = -cx$$

Mathematical Engineering problem.

1. Rectilinear motion. When a body
 moves in a straight line, the
 motion is called Rectilinear motion.
 If x be the distⁿ of body at
 any time t from starting point.
 then its linear velocity v is given
 by

$$v = \frac{dx}{dt}$$

acceleration of the body

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

If m g a body is m & its body
 is moving with the acceleration
 a by the force F acting on it

$$3x^{100} = 21x^{\frac{100}{3}} = 7x^{\frac{100}{3}}$$

$$3x^{100} = 21x^{\frac{100}{3}} = 7x^{\frac{100}{3}}$$